
Week 34: Probability theory

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These notes are written by myself, and errors may and will occur. When in doubt, trust the book and Gunners lectures!

1. Theory based on Karr (1993)

Informally, a *random experiment* has the following objects:

1. Sample space: Set Ω with possible outcomes $\omega \in \Omega$.
2. Events: Subsets of Ω for which probabilities are defined.
3. Measurable space: Sample space Ω equipped with a family of events $\mathcal{E}$.

Definition 1.1. (*σ -algebra*) A σ -algebra $\mathcal{E}$ on Ω is a family of subsets of Ω satisfying

- $\Omega \in \mathcal{E}$
- If $A \in \mathcal{E}$, then $A^C = \Omega \setminus A \in \mathcal{E}$.
- If $A_1, A_2, \dots \in \mathcal{E}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{E}$.

Definition 1.2. (*Events*) The family of events $\mathcal{E}$ is a σ -algebra on Ω .

Definition 1.3. (*Probability*) A probability on $(\Omega, \mathcal{E})$ is a function $P : \mathcal{E} \rightarrow \mathbb{R}$ such that

- $P(A) \geq 0$ for all $A \in \mathcal{E}$.
- $P(\Omega) = 1$.
- If $A_1, A_2, \dots \in \mathcal{E}$ are pairwise disjoint, then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

We often call the tuple $(\Omega, \mathcal{E}, P)$ a *probability space*.

As a simple example, consider a coin toss. Here $\Omega = \{H, T\}$ and $\mathcal{E} = \{\emptyset, \Omega, \{H\}, \{T\}\}$. For a fair coin, we can define a probability with $P(\emptyset) = 0$, $P(\{H\}) = P(\{T\}) = 1/2$, $P(\Omega) = 1$.

Definition 1.4. (*Random variable*) A random variable (RV) is a function $X : \Omega \rightarrow \mathbb{R}$ such that

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{E}$$

for every $B \in \mathcal{B}(\mathbb{R})$. Here $\mathcal{B}(\mathbb{R})$ denotes the Borel σ -algebra on $\mathbb{R}$, which is the smallest σ -algebra containing all open intervals in $\mathbb{R}$ (Karr, 1993).

Going back to our example, we select an RV $X : \Omega \rightarrow \mathbb{R}$ with $X(H) = 1$ and $X(T) = 0$. For any $B \in \mathcal{B}(\mathbb{R})$, we see that

$$X^{-1}(B) = \begin{cases} \emptyset & \text{if } 0 \notin B \text{ and } 1 \notin B \\ \{H\} & \text{if } 0 \notin B \text{ and } 1 \in B \\ \{T\} & \text{if } 0 \in B \text{ and } 1 \notin B \\ \Omega & \text{if } 0 \in B \text{ and } 1 \in B \end{cases}$$

so X is indeed a RV.

Definition 1.5. (*Simple RV*) A RV X is simple if it takes on only finitely many values. We can write this as $X(\Omega) = \{X(\omega) \mid \omega \in \Omega\} = \{x_1, \dots, x_n\}$ for a finite n . Defining $A_i = \{\omega \in \Omega : X(\omega) = x_i\}$, we have that $A_1, \dots, A_n$ is a disjoint partition of Ω , and X can be expressed as $X = \sum_{i=1}^n x_i \mathbb{1}_{A_i}$.

Definition 1.6. (*Expectation of simple RVs*) The expectation of a simple RV $X = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ is defined as $\mathbb{E}[X] = \sum_{i=1}^n a_i P(A_i)$.

2. Problems

Problem 2.1. (*Measurable spaces and probabilities*) Let $\Omega = \{1, 2, 3\}$ and $\mathcal{E} = \{\emptyset, \Omega, \{1\}, \{2, 3\}\}$.

1. Show that $(\Omega, \mathcal{E})$ is a measurable space.
2. How many subsets of Ω are events? Can you give an example of a set that is not an event?
3. Give an example of a probability P on $(\Omega, \mathcal{E})$.

Solution.

1. Need to show that $\mathcal{E}$ is a σ -algebra on Ω . All the conditions listed in [Definition 1.1](#) are easy to verify.
2. There are four events, as listed in $\mathcal{E}$. For an examples of a set that is not an event, we can consider $\{1, 2\} \subset \Omega$. This is not an event since $\{1, 2\} \notin \mathcal{E}$.
3. One example is

$$P(\emptyset) = 0, \quad P(\{1\}) = 1/3, \quad P(\{2, 3\}) = 2/3, \quad P(\Omega) = 1.$$

It is easy to verify that this function satisfies the conditions of a probability.

Problem 2.2. (Random variables) Let $X : \Omega \rightarrow \mathbb{R}$ be such that $(X \leq q)$ is an event for all $q \in \mathbb{Q}$.

1. Prove that $X^{-1}((-\infty, x]) = (X \leq x)$ is an event for every $x \in \mathbb{R}$.
2. Prove that $X^{-1}(I)$ is an event for every interval $I \subseteq \mathbb{R}$.
3. Define $\mathcal{M}_X := \{A \subseteq \mathbb{R} : X^{-1}(A) \in \mathcal{E}\}$, and show that $\mathcal{M}_X$ is a σ -algebra on $\mathbb{R}$.
4. Use the preceding parts to prove that X is a RV.

Solution.

1. There is a rational number between any two different real numbers. Choose $q_n \in \mathbb{Q}$ for every $n \in \mathbb{N}$ such that

$$x < q_n < x + \frac{1}{n}.$$

If $X \leq x$ then $X \leq q_n$. If $X > x$ then eventually $q_n < x + 1/n < X$. Therefore we have

$$(X \leq x) = \bigcap_{n=1}^{\infty} (X \leq q_n). \quad (1)$$

Since $\bigcap_{n=1}^{\infty} (X \leq q_n) = \left(\bigcup_{n=1}^{\infty} (X \leq q_n)^C \right)^C \in \mathcal{E}$, the σ -algebra is closed under countable intersections, giving that $(X \leq x)$ is an event.

2. We also have that strict inequalities are events. This is because

$$(X < x) = \bigcup_{\substack{q \in \mathbb{Q} \\ q < x}} (X \leq q).$$

which is a countable union of events. Complements give

$$(X > a) = (X \leq a)^C, \quad (X \geq a) = (X < a)^C,$$

meaning these are also events. Any interval and singleton can be made by combining and intersecting sets of these four types. For example,

$$(a < X \leq b) = (X > a) \cap (X \leq b),$$

meaning $X^{-1}(I) = (X \in I)$ is an event whenever I is an interval.

3. First, $\mathbb{R} \in \mathcal{M}_X$ because $X^{-1}(\mathbb{R}) = \Omega \in \mathcal{E}$. If $A \in \mathcal{M}_X$, then

$$X^{-1}(A^C) = X^{-1}(\mathbb{R} \setminus A) = \Omega \setminus X^{-1}(A) = (X^{-1}(A))^C \in \mathcal{E},$$

so $A^C \in \mathcal{M}_X$. Finally, if $A_1, A_2, \dots \in \mathcal{M}_X$ then

$$X^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) = \bigcup_{n=1}^{\infty} X^{-1}(A_n) \in \mathcal{E}.$$

Therefore $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}_X$, and $\mathcal{M}_X$ is a σ -algebra on $\mathbb{R}$.

4. Need to show that $X^{-1}(B) \in \mathcal{E}$ for every $B \in \mathcal{B}(\mathbb{R})$. Let $\mathcal{J}$ be the family of open intervals in $\mathbb{R}$. Part 2 shows that $\mathcal{J} \subseteq \mathcal{M}_X$. Since the Borel σ -algebra $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra containing all open intervals, we have

$$\mathcal{B}(\mathbb{R}) \subseteq \mathcal{M}_X.$$

Consequently, $X^{-1}(B) \in \mathcal{E}$ for every $B \in \mathcal{B}(\mathbb{R})$, which proves that X is a RV.

Problem 2.3. (*Probability associated with RV*) Let X be a RV. Show that the function $P_X : \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by

$$P_X(B) := P(X \in B) = P(X^{-1}(B))$$

is a probability on $\mathbb{R}$.

Solution. For P_X to define a probability distribution on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we simply check the three conditions:

- With $B \in \mathcal{B}(\mathbb{R})$ we have $P_X(B) = P(X \in B) \geq 0$ since $P(\cdot)$ is a probability.
- $P_X(\mathbb{R}) = P(X^{-1}(\mathbb{R})) = P(\Omega) = 1$.
- With pairwise disjoint $B_1, B_2, \dots \in \mathcal{B}(\mathbb{R})$, we have

$$P_X\left(\bigcup_{i=1}^{\infty} B_i\right) = P\left(X^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right)\right) = P\left(\bigcup_{i=1}^{\infty} X^{-1}(B_i)\right) = \sum_{i=1}^{\infty} P(X^{-1}(B_i)) = \sum_{i=1}^{\infty} P_X(B_i).$$

Problem 2.4. (Proving some important properties of expectation for simple RVs) Let X and Y be simple RVs, and let $\alpha, \beta \in \mathbb{R}$. Prove

- Relation to probability: If $A \in \mathcal{E}$, then $\mathbb{E}[\mathbb{1}(A)] = P(A)$.
- Constants are preserved: If $X = c$ for some $c \in \mathbb{R}$, then $\mathbb{E}[X] = c$.
- Monotonicity: If $X \leq Y$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.
- Linearity: $\mathbb{E}[\alpha X + \beta Y] = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y]$.

Solution. Denote the simple RVs as $X = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ and $Y = \sum_{j=1}^m b_j \mathbb{1}_{B_j}$.

- As $\mathbb{1}_A = 1 \cdot \mathbb{1}_A + 0 \cdot \mathbb{1}_{A^C}$, the indicator function is clearly simple. This means

$$\mathbb{E}[\mathbb{1}(A)] = 1 \cdot P(A) + 0 \cdot P(A^C) = P(A).$$

- If $X(\omega) = c \in \mathbb{R}$ for every $\omega \in \Omega$, then $\mathbb{E}[X] = \mathbb{E}[c \mathbb{1}_\Omega] = cP(\Omega) = c$.
- If $X(\omega) \leq Y(\omega)$ for all $\omega \in \Omega$, then $a_i \leq b_j$ whenever $A_i \cap B_j \neq \emptyset$. Since both A_i and B_j give a disjoint partitions of Ω , we have

$$\mathbb{E}[X] = \sum_{i=1}^n a_i P(A_i) = \sum_{i=1}^n \sum_{j=1}^m a_i P(A_i \cap B_j) \leq \sum_{i=1}^n \sum_{j=1}^m b_j P(A_i \cap B_j) = \sum_{j=1}^m b_j P(B_j) = \mathbb{E}[Y].$$

- We can express

$$\alpha X + \beta Y = \sum_{i=1}^n \sum_{j=1}^m (\alpha a_i + \beta b_j) \mathbb{1}_{A_i \cap B_j},$$

meaning that $\alpha X + \beta Y$ is also simple. Thus

$$\begin{aligned} \mathbb{E}[\alpha X + \beta Y] &= \sum_{i=1}^n \sum_{j=1}^m (\alpha a_i + \beta b_j) P(A_i \cap B_j) \\ &= \alpha \sum_{i=1}^n \sum_{j=1}^m a_i P(A_i \cap B_j) + \beta \sum_{i=1}^n \sum_{j=1}^m b_j P(A_i \cap B_j) \\ &= \alpha \sum_{i=1}^n a_i P(A_i) + \beta \sum_{j=1}^m b_j P(B_j) \\ &= \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y]. \end{aligned}$$

Proposition 2.1. *(A more general definition of the expectation) A more general definition of expectation is using the Lebesgue integral*

$$\mathbb{E}[X] = \int_{\Omega} X \, dP.$$

All the details of this is likely beyond the scope of this course. However, we can still try to gain a bit of insight. For a RV $X : \Omega \rightarrow \mathbb{R}$, we can rewrite this as an integral over $\mathbb{R}$ using the induced probability measure P_X

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) \, dP(\omega) = \int_{\mathbb{R}} x \, dP_X(x).$$

For a simple RV $X = \sum_{i=1}^n x_i \mathbb{1}_{A_i}$ the distribution is discrete $P_X(\{x_i\}) = P(A_i)$, so

$$\int_{\mathbb{R}} x \, dP_X(x) = \sum_{i=1}^n x_i P_X(\{x_i\}) = \sum_{i=1}^n x_i P(A_i)$$

becomes the same as [Definition 1.6](#). If the distribution instead is continuous with a probability density function f_X , then $dP_X(x) = f_X(x) \, dx$. For the dx we can use the standard Riemann integral. This gives the familiar expression

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x) \, dx,$$

that you might remember from the first course in statistics.

References

Karr, A. F. *Probability*. Springer-Verlag, 1993. <https://doi.org/10.1007/978-1-4612-0891-4>